A/ Common Discrete Probability distributions

X is a discrete random variable defined on a probability space $(\Omega, \Lambda, \mathrm{P})$.

1. Uniform probability distribution

Definition 1:
Let $N \in IN^*$, X follows a uniform probability distribution over the set $X(\Omega) = \{1,2,...,N\}$ if:

$$\forall k,\ 1 \le k \le N,\ \mathrm{P}_X(k) = \frac{1}{N}.$$

We denote it as $X \to U_{\{1,2,..,N\}}$ and we do have: $\sum_{k=1}^{N} \mathrm{P}_X(k) = 1$

Property 1: if $X \to U_{\{1,2,..,N\}}$ where $N \ge 1$, then : $\begin{cases} E(X) = \dfrac{N+1}{2} \\ V(X) = \dfrac{N^2-1}{12} \end{cases}$

(Apart from the use of generating functions (see Ch4), we use those formulas,

$\sum_{1}^{N} k = \frac{N(N+1)}{2}$ and $\sum_{1}^{N} k^2 = \frac{N(N+1)(2N+1)}{6}$ to derive $E(X)$ and $V(X)$ using their definitions.)

Example 1: A fair die is rolled. Let X be the r.v. representing the number on the face obtained.

$X \to U_{\{1,2,..,6\}}$ and we have : $E(X) = \frac{7}{2}$ et $V(X) = \frac{35}{12}$

2. Bernoulli distribution

Definition 2: Bernoulli Trial
Consider an experiment where the sample space contains only two elementary events, $\Omega = \{A, \overline{A}\}$.
We define:

- "Success" as the occurrence of event A .
- "Failure" as the occurrence of event $\overline{A}$.

Setting: $\mathrm{P}(A) = p$ et $\mathrm{P}(\overline{A}) = q = 1 - p$

When an event and its complement are of interest, the experiment is called a **Bernoulli trial**.

Example 2: a player flips a biased coin that lands on heads with probability¾ and tails with probability ¼

Let A be the event: « the outcome is heads », we have $\mathrm{P}(A) = \frac{1}{4}$ and $\mathrm{P}(\overline{A}) = \frac{3}{4}$

Definition 3:
Let $p \in IR$, $0 \le p \le 1$. X follows a **Bernoulli distribution** with parameter p, denoted: $X \to \mathrm{B}(p)$ if:
$X(\Omega) = \{0,1\}$ such as $\mathrm{P}_X(1) = p$ and $\mathrm{P}_X(0) = 1 - p$

Note that we have: $\sum_{k \in X\Omega)} \mathrm{P}_X(k) = 1$

Property 2: If $X \to \mathrm{B}(p)$, then $E(X) = p$ et $V(X) = p(1-p)$

Example 3: If $X \to \mathrm{B}\left(\frac{1}{4}\right)$, then: $E(X) = \frac{1}{4}$ and $V(X) = \frac{3}{16}$.

Exercise 1: One draw is made from an urn containing two categories of balls;

- White balls in proportion p.
- Black balls in proportion $q = 1 - p$.
  Let X be the number of white balls drawn. Find the distribution, expectation, and variance of X

3. Binomial Distribution

Consider a N sequence of independent and identical Bernoulli trials.
In the ith trial, we define the r.v.: $X_i \to \mathrm{B}(p)$
Let X be the total number of successes in the N trials:
Then $X = \sum_{i=1}^{N} X_i$ follows a **binomial distribution**.

Definition 4:
Let $N \in IN^*$ and $p \in [0,1]$.
X follows a binomial distribution with parameters N and p, that we note $X \to \mathrm{B}(N; p)$ if:

$$X(\Omega) = \{0,1,\ldots,N\} \quad \text{such as} \quad \forall k \in X(\Omega) \quad , \quad \mathrm{P}_X(k) = C_N^k . p^k . (1-p)^{N-k} \text{ (Justify)}$$

Remark 1: It's easy to show that we have $\sum_{k=0}^{N} \mathrm{P}_X(k) = 1$. It comes from the general term of Newton's binomial theorem, hence the name **binomial** distribution.

Remark 2: The binomial distribution $\mathrm{B}(1; p)$ corresponds to the Bernoulli distribution $\mathrm{B}(p)$.

Property 3:
If $X \to \mathrm{B}(N; p)$, $N \in IN^*$ and $0 \le p \le 1$, then: $E(X) = N.p$ and $V(X) = N.p(1-p)$

Example 4: A biased coin (from Example 2) is flipped 10 times. Let X be the number of heads obtained.
Answer to: $X \to ?$, $E(X) = ?$ and $V(X) = ?$

Property 4: If $X_1 \to \mathrm{B}(N_1; p)$ and $X_2 \to \mathrm{B}(N_2; p)$, and X_1 and X_2 are independent, then:

$$X_1 + X_2 \to \mathrm{B}(N_1 + N_2; p)$$

Indication: to demonstrate this property, we can use the pdf of the sum (see chapter5), in this case we should know this Vandermonde's formula, $C_{N_1+N_2}^{n} = \sum_{k=0}^{n} C_{N_1}^{k} C_{N_2}^{n-k}$. we can also use the property of the generating function of a r.v. "sum" of two independent variables.

Generalisation: let $X_1, X_2, ..., X_n$ be n *i.i.d* $\mathrm{B}(N_i, p)$ R.Vs, then $\sum_{i=1}^{n} X_i \to \mathrm{B}(\sum_{i=1}^{n} N_i; p)$.

Exercise 2: n draws with replacement are made from an urn with:

- White balls in proportion p.
- Black balls in proportion $q = 1 - p$.

  Let X be the number of white balls drawn. Find its distribution, expectation, and variance.

4. Poisson Distribution

Definition 5:

A random variable X follows a **Poisson distribution** with parameter $\lambda \in IR^*$ if:

$$X(\Omega) = IN \quad \text{such as} \quad \forall k \in IN \quad , \quad \mathrm{P}_X(k) = e^{-\lambda} \cdot \frac{\lambda^k}{k!}$$

We denote it as $X \to Poisson(\lambda)$

Property 5: If $X \to Poisson(\lambda), \lambda > 0$, then: $E(X) = \lambda$ and $V(X) = \lambda$

Property 6: If $X_1 \to Poisson(\lambda_1)$ and $X_2 \to P(\lambda_2)$, and X_1 and X_2 are independent, then:

$$X_1 + X_2 \to Poisson(\lambda_1 + \lambda_2)$$

Indication : to demonstrate this property, we can use the pdf of the sum (see chapter5) by considering in the calculus $\sum_{k=0}^{n} \frac{n!}{k!(n-k)!} \lambda_1^k \lambda_2^{n-k} = \sum_{k=0}^{n} C_n^k \lambda_1^k \lambda_2^{n-k} = (\lambda_1 + \lambda_2)^n$. Or we can also use the property of the generating function of a r.v. "sum" of two independent variables.

Generalisation : let $X_1, X_2, ..., X_n$ be n $Poisson(\lambda_i)$ r.Vs, then : $\sum_{i=1}^{n} X_i \to P(\sum_{i=1}^{n} \lambda_i)$

5. Approximation:

If X follows $\mathrm{B}(N; p)$ with N large and p small (close to 0), then the CDF of X (i.e. $\mathrm{B}(N; p)$) can be approximated by the CDF of Poisson distribution, $Poisson(\lambda)$, with $\lambda = N.p$.

Remark 4: In practice, this approximation is valid for $N \geq 30$, $p \leq 0.1$ and $N.p < 15$, or $N > 50$, $p < 0.1$.

In theory, by considering the approximation conditions:

$N \to \infty$, $p \to 0$ and $N.p$ being finite, we show that:

$\lim\limits_{N \to +\infty} C_N^k p^k (1-p)^{N-k} = e^{-\lambda} \frac{\lambda^k}{k!}$, with $\lambda = \lim\limits_{N \to +\infty} N.p(N)$, p is a function of N.

We usually write $X \xrightarrow{Loi} P(\lambda)$ (See Chapter 2 in course of PRST2).

Remark 5: The Poisson distribution is used to model counting experiments of rare events, such as:

- The number of calls received by a telephone switchboard.
- The number of passengers arriving at a ticket counter.

If events occur independently and only one event happens in a short time interval Δt with probability $\lambda.\Delta t$, we can approach the probability distribution of the number of events occurring in a given period by a $Poisson(\lambda)$.

6. Hypergeometric Distribution:

Definition 6:
We perform n draws *without replacement* from an urn containing two categories of balls: N_1 white balls and N_2 black balls. We set $N = N_1 + N_2$ as the total number of balls.
Let X be the number of white balls drawn:

- Possible values of X : All values between $\max(0, n - N_2)$ and $\min(N_1, n)$.

- Probabilities: $\forall k \in X(\Omega), \quad \mathrm{P}(X = k) = \dfrac{C_{N_1}^{k} \cdot C_{N_2}^{n-k}}{C_N^n}$

By setting $p = \dfrac{N_1}{N}$ as the proportion of white balls, $q = \dfrac{N_2}{N}$ as the number of black balls, and n as the number of draws, we have: $\begin{cases} E(X) = n.\dfrac{N_1}{N} = n.p \\ V(X) = n.p.q.\dfrac{N-n}{N-1} \end{cases}$

We denote this as $X \to H(N, n, p)$.

Remark 7:
The hypergeometric distribution $H(N, n, p)$ can be approximated by the binomial distribution $\mathrm{B}(n, p)$ when $N_1 \to \infty$, $N_2 \to \infty$ and with $\dfrac{N_1}{N}$ and $\dfrac{N_2}{N}$ remaining finite.

In practice, this approximation is valid when $\dfrac{n}{N} < 0.1$.

7. Pascal Distribution

Definition 7:
We perform draws *with replacement* from an urn containing two categories of balls: white balls in proportion p

and black balls in proportion $q = 1 - p$.
Let X be the number of draws needed to obtain r white balls ("successes"):

- Possible values of X : $X(\Omega) = \{r, r+1, ..., k, ...\}$
- Probability mass function: $\forall k \in X(\Omega),\ \mathrm{P}_X(k) = \mathrm{P}(X = k) \quad = \quad C_{k-1}^{r-1}.p^r.q^{k-r}$

We denote this as $X \to Pascal(r, p)$ and we have: $\begin{cases} E(X) = \dfrac{r}{p} \\ V(X) = \dfrac{r.q}{p^2} \end{cases}$

7.1) Geometric Distribution:

If $r = 1$, the Pascal distribution is called the geometric distribution, which corresponds to drawing with replacement until the first white ball appears (first success).

We denote this as $X \to \mathcal{G}(p)$, and we have: $\begin{cases} \forall k \in IN^*, \mathrm{P}_X(k) = p.q^{k-1}, \\ E(X) = \dfrac{1}{p} \quad and \quad v(X) = \dfrac{q}{p^2} \end{cases}$

Moments Generating Function:
If $X \to \mathcal{G}(p)$, then for $t \in IR$, its generating function is given by:

$$\begin{aligned} G_X(t) = E(e^{t.X}) &= \sum_{k \in (\Omega)} e^{t.k} \mathrm{P}_X(k) \\ &= \sum_{k=1}^{+\infty} e^{t.k}.p.q^{k-1} \\ &= p.e^t \sum_{k=1}^{+\infty} (q.e^t)^{k-1} \\ &= \frac{p.e^t}{1 - q.e^t}, \qquad \left|q.e^t\right| < 1 \end{aligned}$$

We conclude that : $X \to \mathcal{G}(p),\ \forall t \in IR,$ such as $0 < qe^t < 1$, then $G_X(t) = \dfrac{p.e^t}{1-(1-p).e^t}$

Remark 8:

For $X \to Pascal(r, p)$, considering the property that $X = \sum_{i=1}^{r} X_i$ such as $X_1, X_2, ...X_r \quad i.i.d\ \mathcal{G}(p)$,

we easily deduce the generating function of X by using a property of moments generating function:

$$t \in IR \text{ such as } 0 < qe^t < 1 \text{ we have : } G_X(t) = \left(\frac{p.e^t}{1-(1-p).e^t}\right)^r.$$

Property 7: If $X_1 \to Pascal(r_1; p)$ and $X_2 \to Pascal(r_2; p)$ are independent, then:
$X_1 + X_2 \to Pascal(r_1 + r_2; p)$

Generalization: $X_1, X_2, ..., X_n$ n $i.i.d.$ $Pascal(r_i, p)$ r.Vs , then: $\sum_{i=1}^{n} X_i \to Pascal(\sum_{i=1}^{n} r_i; p)$

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7.2) Negative Binomial Distribution:

We perform draws *with replacement* from an urn containing two categories of balls: white balls in proportion p and black balls in proportion q.
Let Y be the number of draws needed to obtain r white balls ("successes"). The random variable $X = Y - r$ follows the negative binomial distribution with parameters r and p .We denote this as $X \to BN(r, p)$.

Exercise 3: Show that if $X \to BN(r, p)$, then $\begin{cases} \forall k \in IN, \quad P_X(k) = C_{r+k-1}^{r-1} p^r . q^k \\ \qquad et \\ E(X) = \dfrac{r.q}{p} \quad et \quad v(X) = \dfrac{r.q}{p^2} \end{cases}$

Exercise 4: Find the generating function of the random variable $X \to BN(r, p)$.

B/ Common Continuous Probability Distributions:

8. Uniform Distribution:

Definition 8:
A continuous random variable X follows the uniform distribution over the real interval $[a, b]$ if it has the probability density function f defined from IR to IR as follows:

$$f(x) = \begin{cases} \dfrac{1}{b-a} & si\ x \in [a,b] \\ 0 & \sin on \end{cases}$$

We denote it as $X \to U(a,b)$, and we have: $\begin{cases} E(X) = \dfrac{b+a}{2} \\ V(X) = \dfrac{(b-a)^2}{12} \end{cases}$ and $F_X(x) = \begin{cases} 0 & si & x < a \\ \dfrac{x-a}{b-a} & si & a \le x \le b \\ 1 & si & x > b \end{cases}$

Remark 9:
The uniform distribution is used to model a random variable that takes values randomly over an interval $[a, b]$.
$U(0,1)$ corresponds to the generators of "random numbers," and we can produce random numbers over $[a, b]$ using the transformation $y = (b-a)x + a$, (see Exercise 8 in the tutorial).
It also allows simulating any other distribution using "random numbers" (see Exercise 9 in the tutorial).

9. Exponential Distribution:

Definition 9:

A continuous random variable X follows the exponential distribution with parameter λ, $(\lambda > 0)$, if it has the probability density function f defined from f to f as follows: $f(x) = \begin{cases} \lambda.e^{-\lambda.x} & si\ x \geq 0 \\ 0 & \sin on \end{cases}$

We denote it as $X \to \mathrm{E}xp(\lambda)$, and we have: $\begin{cases} E(X) = \dfrac{1}{\lambda} \\ V(X) = \dfrac{1}{\lambda^2} \end{cases}$ *and* $F_X(x) = \left(1 - e^{-\lambda.x}\right) I_{IR^+}(x)$

Remark 10:
The exponential distribution is used to measure waiting times, service times, … , etc.
It is the continuous counterpart of the geometric distribution.
It corresponds to the random variable representing the time elapsed between two successive occurrences in a "Poisson process," where λ is the average number of occurrences per unit of time, and $1/\lambda$ is the mean duration between two occurrences.
The exponential distribution serves as a model for the lifetime of a system ("Reliability"), where $1/\lambda$ represents the expected lifetime of the system.

10. Gamma Distribution:

Definition 10:
A continuous random variable X follows the gamma distribution with parameters p and, denoted as $\Gamma(p,\theta)$, if its probability density function is:

$$f(x) = \frac{\theta^p}{\Gamma(p)} e^{-\theta.x} x^{p-1} I_{IR^+}(x), \quad p > 0, \theta > 0 \qquad \text{where} \quad \Gamma(p) = \int_0^{+\infty} e^{-x} x^{p-1} dx \ .$$

$\Gamma(p)$ is known the Euler's **gamma function**.

- If $\theta = 1$, we denote $\Gamma(p,1) = \gamma(p)$, with the density function: $f(x) = \dfrac{1}{\Gamma(p)} e^{-x} x^{p-1} I_{IR^+}(x), \quad p > 0$.
- **Exercise 5** (in tutorial sessions): If $\theta \neq 1$, we can show that $Y = \theta.X \to \gamma(p)$.
- If $p = 1$, we retrieve the **exponential distribution** with parameter $\theta = 1$.

Properties of the Euler's Gamma Function

Euler's gamma function $\Gamma(p) = \int_0^{+\infty} e^{-x} x^{p-1} dx$ satisfies the recurrence relation:

$$\begin{cases} \Gamma(p) = (p-1).\Gamma(p-1), \quad p > 0 \\ \Gamma(p) = (p-1)!, \quad p \in IN^* \\ \Gamma(\tfrac{1}{2}) = \sqrt{\pi} \end{cases} .$$

Moments: If $X \to \Gamma(p,\theta)$, then: $M_k(X) = E(X^k) = \dfrac{\Gamma(p+k)}{\theta^k.\Gamma(p)}$, and we deduce: $\begin{cases} E(X) = \dfrac{p}{\theta} \\ et \\ V(X) = \dfrac{p}{\theta^2} \end{cases}$.

Theorem 1: Let X and Y be two independent gamma-distributed random variables such as they follow $\Gamma(p_1,\theta)$ and $\Gamma(p_2;\theta)$ respectively. Then: $X+Y \to \Gamma(p_1+p_2;\theta)$

Generalization: Let $X_1, X_2,..., X_n$ n be independent gamma-distributed random variables such that: $\forall i = 1,2,...,n,\quad X_i \to \Gamma(p_i,\theta)$. Then their sum follows a gamma distribution: $X = \sum_{i=1}^{n} X_i \to \Gamma\left(\sum_{i=1}^{n} p_i;\theta\right)$

Remark 12: The gamma distribution models the waiting time of one occurrence from the next r^{th} in a Poisson process. It plays a similar role to the negative binomial distribution in a Bernoulli process.
Its cumulative distribution function (CDF) is not explicit and typically requires numerical computations (software tools) since gamma distribution tables are not common.

Moment-Generating Function (MGF):

$X \to \Gamma(p,\theta)$, its moment-generating function is given by: for $t \in IR$,

$$G_X(t) = E(e^{t.X}) = \int_{IR} e^{t.x} f_X(x)dx = \frac{\theta^p}{(\theta-t)^p.\Gamma(p)} \int_0^{+\infty} e^{-y} y^{p-1} dy = \frac{\Gamma(p)}{\Gamma(p)}.\left(\frac{\theta}{\theta-t}\right)^p = \left(\frac{\theta}{\theta-t}\right)^p$$

(by sing the change of variable $y = (\theta-t)x$)
we obtain the domain condition $t < \theta$, ensuring the integral's convergence.

11. Normal Distribution

The normal distribution is one of the most important probability distributions, if not the most important. We will discover its significance as we progress through this PRST1 course and the PRST2 course.
In this chapter, we will familiarize ourselves with some properties of the normal distribution and the most important distributions derived from it.

Definition 11:
A continuous random variable X follows a normal distribution (or Laplace-Gauss distribution) with parameters $\mu \in IR$ and $\sigma \in IR_+^*$ if its the probability density function is given by:

$$\begin{aligned} f : IR &\to IR \\ x &\mapsto f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} \end{aligned}$$

This is denoted as: $X \to N(\mu;\sigma)$

Remark 13:
Some use the notation $X \to N(\mu;\sigma^2)$, so be careful with the chosen notation, especially when the parameters are given numerically. This is to avoid confusion between the standard deviation and the variance, as explained in the following remark.

Remark 14:
For now, we consider μ and σ as two real parameters defined earlier. We will see that they correspond to the mean (expectation) $E(X)$ and the standard deviation $\sigma(X)$, hence the chosen notation.

Moments:

$X \to N(\mu;\sigma)$, it can be shown that:

$$\begin{cases} E(X) = \mu, & \\ \forall k \in IN, & \mu_{2k+1} = \mu_{2k+1}(X) = 0 \quad \textit{centred momentsof evenorder} \\ \forall k \in IN^*, & \mu_{2k} = \mu_{2k}(X) = 1\times 3\times \ldots \times (2k-1)\sigma^{2k} \ \textit{centred momentsof odd order} \end{cases}$$

In particular, we have: $V(X) = \mu_2 = (2\times 1-1)\sigma^{2\times 1} = \sigma^2$.

Property 7: If $X \to N(\mu;\sigma)$, then $\forall (a,b) \in IR^* \times IR, \quad Y = aX + b \to N(a\mu + b; |a|\sigma)$.

Proof: This follows from a special case of Theorem 3 from Chapter 4.

Remark 15:
The mean and variance of $Y = aX + b$ are established in Corollary 1 and Corollary 2 of Chapter 4. The interesting aspect of this property is that an affine transformation of a normally distributed variable remains normally distributed. This is one of the key characteristics of the normal distribution.

Standard Normal Distribution

From its name, we understand that it is a normal distribution whose parameters (i.e., mean and standard deviation) are respectively 0 (centered) and 1 (standardized).

From the previous property: $X \to N(\mu;\sigma)$ then $T = \dfrac{X - E(X)}{\sigma(X)} = \dfrac{X-\mu}{\sigma} \to N(0;1)$

Thus, we define: (in below)

Definition 12:
A continuous random variable follows the standard normal distribution if its probability density function is given by:

$$f_T(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad \forall x \in IR$$

This is often denoted as $T \to N(0,1)$.

Remark 16:

- The density function of the standard normal distribution $N(0;1)$ is often denoted ϕ.

- From the Gauss' integral $\int_{-\infty}^{+\infty} e^{-\frac{t^2}{2}} dt = \sqrt{2\pi}$, this allows us to begin our course by defining the standard normal distribution $N(0;1)$ by its pdf $f_T(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}, \quad \forall x \in IR$, satisfying the three conditions of a probability density.
- The curve of f_T (see the statistical table that will be provided in tutorial) is a "bell-shaped" curve known as the Gaussian curve, which is symmetric about the y-axis.
- The standard normal distribution satisfies: $E(T) = 0$, $V(T) = 1$ and then $\sigma(T) = 1$

<u>Cumulative Distribution Function (CDF)</u>

For $T \to N(0;1)$, the cumulative distribution function is given by:

$$\forall x \in IR, \quad F_T(x) = \mathrm{P}(T \le x) = \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

There is no closed-form expression for the integral of $x \to \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$, (its integral exists, but we do not know its analytic expression), so its values have been approximated numerically and compiled into a statistical table. This table provide values of $F_T(x)$ for positive values of x, and for negative values, symmetry properties are used.

<u>A key property</u>: $T \to N(0;1)$ then: $\forall a \in IR$, $\begin{cases} \Phi(-a) = 1 - \Phi(a) \quad \textit{in particular} \quad \Phi(0) = \frac{1}{2} \\ \mathrm{P}(|T| \le a) = 2.\Phi(t) - 1 \end{cases}$

<u>Probability Calculations for Any Normal Distribution</u>

For a normal variable $X \to N(\mu;\sigma)$, probabilities are computed using the standard normal distribution by making the transformation $T = \frac{X-\mu}{\sigma}$, because we know that T follows the standard normal distribution. Since we only have tables for $N(0;1)$. (We compute probabilities by converting X into T)

For example: $\mathrm{P}(x_1 < X < x_2) = \mathrm{P}\left(\frac{x_1-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{x_2-\mu}{\sigma}\right) = \mathrm{P}(t_1 < T < t_2) = \Phi(t_2) - \Phi(t_1)$

These values $\Phi(t_1)$ and $\Phi(t_2)$ are then determined directly using the standard normal table if t_1 et $_2$ are positives, otherwise we use the key property given previously.

- **<u>Some possible situations</u>:** after converting $X \to N(\mu;\sigma)$ into $T = \frac{X-\mu}{\sigma} \to N(0;1)$, the following situations can arise:

  - $\mathrm{P}(T < t) = \begin{cases} \Phi(t) & si \quad t \ge 0 \\ 1 - \Phi(-t) & si \quad t < 0 \end{cases}$

- $\mathrm{P}(T > t) = \begin{cases} 1-\Phi(t) & si \quad t \geq 0 \\ \Phi(-t) & si \quad t < 0 \end{cases}$
- $\mathrm{P}(t_1 < T < t_2) = \begin{cases} \Phi(t_2)-\Phi(t_1) & si \quad 0 \leq t_1 < t_2 \\ \Phi(t_2)+\Phi(-t_1)-1 & si \quad t_1 < 0 \leq t_2 \\ \Phi(-t_1)-\Phi(-t_2) & si \quad t_1 < t_2 \leq 0 \end{cases}$
- $\mathrm{P}(|T| < t) = 2\Phi(t)-1$
- $\mathrm{P}(|T| > t) = 2(1-\Phi(t))$

<u>Some Key Values of the Normal Distribution</u>

From the statistical table of $N(0;1)$, we find: $\Phi(1.96) \approx 0.975$, $\Phi(2.57) \approx 2.57$ et $\Phi(3.3) \approx 0.999$, this leads to the following dispersion intervals around the mean $X \to N(\mu;\sigma)$:

$$\mathrm{P}(\mu - 1.96 \times \sigma < X < \mu + 1.96 \times \sigma) = 0.975,$$

$$\mathrm{P}(\mu - 2.57 \times \sigma < X < \mu + 2.57 \times \sigma) = 0.995,$$

$$\mathrm{P}(\mu - 3.30 \times \sigma < X < \mu + 3.30 \times \sigma) = 0.999.$$

The first relation is often summarized as saying that "the probability of obtaining a value within two standard deviations of the mean is 97.5%." This property is sometimes mistakenly applied to other types of distributions. The third relation shows that values beyond three standard deviations from the mean are extremely rare.

<u>Theorem 2</u>: (Sum of Independent Normal Variables)

Let X and Y be two <u>independent</u> normal random variables, $X \to N(\mu_1;\sigma_1)$ and $Y \to N(\mu_2;\sigma_2)$.
Then: $X + Y \to N(\mu_1 + \mu_2; \sqrt{\sigma_1^2 + \sigma_2^2})$

<u>Generalization</u>: If $X_1, X_2, ..., X_n$ are n independent normal variables with:

$$\forall i = 1,2,...,n \quad X_i \to N(\mu_i;\sigma_i)$$

then their sum X follows: $X = \sum_{i=1}^{n} X_i \to N\left(\sum_{i=1}^{n} \mu_i; \sqrt{\sum_{i=1}^{n} \sigma_i^2}\right)$.

<u>Remark17</u>: (This result is particularly important for Chapter 2 in PRST2.)

<u>Moment-Generating Function (MGF)</u>:

$X \to N(0;1)$, its moment-generating function is given by: for $t \in IR$,

$$G_X(t) = E(e^{t.X}) = \int_{IR} e^{t.x} f_X(x)dx$$
$$= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{t.x} e^{-\frac{x^2}{2}} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \exp\left\{-\frac{(x-t)^2}{2} + \frac{t^2}{2}\right\} dx$$
$$= e^{\frac{t^2}{2}} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{(x-t)^2}{2}} dx = e^{\frac{t^2}{2}} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{y^2}{2}} dy, \qquad y = x - t$$
$$= e^{\frac{t^2}{2}}$$

We therefore note: $X \to N(0;1)$, $G_X(t) = e^{\frac{t^2}{2}}$, $t \in IR$.

Exercise6 : (n⁰ 13 in worsheet). Given the moments generating function of $Z \to N(0;1)$, show that the moments generating function of $X \to N(\mu;\sigma)$ is : $G_X(t) = e^{\frac{\sigma^2.t^2}{2} + \mu.t}$, $t \in IR$

12. Log-Normal Distribution

Definition 13:
A continuous positive random variable X follows a log-normal distribution with parameters μ and σ, $(\mu \in IR, \sigma \in IR_+^*)$ if and only if its natural logarithm follows a normal distribution with the same parameters.

The probability density function of X is given by: $f(x) = \begin{cases} \frac{1}{x.\sigma.\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{\log x - \mu}{\sigma}\right)^2} & si\ x > 0 \\ 0 & \sin on \end{cases}$

(We use the change variable theorem seen in chapter4 to calculate f)

Remark 18: The standard normal table can be used to compute probabilities without needing the density function of a log-normal distribution using the equivalence relation:

$$X \to \ell N(\mu;\sigma) \Leftrightarrow T = Log(X) \to N(\mu,\sigma)$$

The log-normal distribution is widely used in biology, economics, and physics, as it models strictly positive variables with a skewed distribution extending towards higher values.

Remark 19: From the equivalence relation in Remark18, we can write:

$$E(X^k) = E(e^{k.T}) = G_T(k), \qquad T \to N(\mu;\sigma)$$

Then given the moments generating function of $N(\mu;\sigma)$, allows us to retrieve $E(X)$, $E(X^2)$ and $V(X)$ by replacing $k = 1$ then $k = 2$ in $G_T(k)$.

C/ Derived Probability Distributions from the Normal Distribution:

13. Chi-Square Distribution:

Definition 14: Let there $(X_1,...,X_n)$ be a system of n independent random variables, such that:

$$\forall i \in \{1,2,...,n\} \qquad X_i \to N(0,1)$$

Then $\sum_{i=1}^{n} X_i^2$ follows a Chi-square distribution with degrees of freedom n, denoted as $\chi^2(n)$.
This distribution plays an important role in hypothesis testing theory and is tabulated.

Remarks 20:

- If $X \to \chi^2(n)$, for $n > 30$, then $\sqrt{2X} - \sqrt{2n-1}$ follows approximately a distribution $N(0,1)$. (Fisher's approximation)
  Thus, for $n > 30$, the quantile of order α of the distribution $\chi^2(n)$ is:

$$\chi_\alpha^2(n) \approx \frac{1}{2}(t_\alpha + \sqrt{2n-1})^2$$

  where t_α is the quantile of order α of the normal distribution $N(0,1)$.

- **Exercise 7:** (in tutorial 6). By proving the following result,

$$T \to N(0;1), \text{ then } \frac{1}{2}T^2 \to \Gamma(\tfrac{1}{2},1) = \gamma(\tfrac{1}{2}).$$

  Based on the generalization of **Theorem 2**, we deduce that if $T_1, T_2, ..., T_n$ n r.v. $i.i.d.$ $N(0;1)$, then: $\frac{1}{2}\sum_{i=1}^{n} T_i^2 = \frac{1}{2}X \to \Gamma(\tfrac{n}{2},1) = \gamma(\tfrac{n}{2})$

  According to **Exercise 5** in **Remark 11**: $X = \sum_{i=1}^{n} T_i^2 \to \Gamma(\tfrac{n}{2}, \tfrac{1}{2})$

  (which gives the probability density function of the distribution $\chi^2(n)$)

  Question: Develop the steps of **Exercise 7** to derive the density of a distribution $\chi^2(n)$.

  Conclusion: We also conclude that $\chi^2(n) = \Gamma(\tfrac{n}{2}, \tfrac{1}{2})$, and consequently,

$$G_X(t) = \left(\frac{1}{1-2t}\right)^{\frac{n}{2}}, \quad t < \tfrac{1}{2}.$$

Moments: $X \to \chi^2(n)$, from the relationship $\chi^2(n) = \Gamma(\tfrac{n}{2}, \tfrac{1}{2}) = \gamma(\tfrac{n}{2})$, we have then:

$$M_k(X) = E(X^k) = 2^k . \frac{\Gamma(\tfrac{n}{2}+k)}{\Gamma(\tfrac{n}{2})}$$

Hence, $\begin{cases} E(X) = n \\ V(X) = 2.n \end{cases}$

Theorem 3: Let X and Y be two independent r.v. following respectively the distributions $\chi^2(n)$ and $\chi^2(m)$. Then, we show that the r.v. $X + Y \to \chi^2(n+m)$.

Generalization: Let $X_1, X_2, ..., X_n$ n independent r.v. such that:

$$\forall i = 1,2,...,n \quad X_i \to \chi^2(n_i)$$

Then the r.v. $X = \sum_{i=1}^{n} X_i \to \chi^2(\sum_{i=1}^{n} n_i)$.

14. Student's Distribution:

Definition 15: If X follows $N(0,1)$ and Y follows $\chi^2(n)$, and if X and Y are independent, then the r.v. $T = \frac{X}{\sqrt{Y/n}}$ follows a Student's distribution with n degrees of freedom, denoted as $St(n)$.
This distribution is used in estimation and hypothesis testing and is tabulated.

We have: $E(T) = 0$ for $n > 1$ and $V(T) = \frac{n}{n-2}$ for $n > 2$.

Remark 21: For $n > 30$, the Student's distribution can be approximated by the standard normal distribution.

Special Case:
For $n = 1$, the r.v. $T = \frac{X}{\sqrt{Y}}$, where X and Y are independent and follow respectively $N(0;1)$ and $\chi^2(1)$, is a well-known continuous distribution called the **Cauchy distribution** (which is also the distribution of the ratio of two independent standard normal r.vs).

The density of T in this case is: $f(x) = \frac{1}{\pi}.\frac{1}{1+x^2}, \forall x \in IR.$

15. Fisher-Snedecor Distribution:

Definition 16: If $X \to \chi^2(m)$ and $Y \to \chi^2(n)$, and if X and Y are independent, then the r.v. $\frac{X/m}{Y/n}$ follows a **Fisher-Snedecor distribution** with m and n degrees of freedom, denoted as $F(m,n)$.

Remark 22:

- This distribution is also used in hypothesis testing and is tabulated.

- $X \to F(m,n)$,

  We have: $E(X) = \dfrac{n}{n-2}$ for $n > 2$ and $V(X) = \dfrac{2n^2(m+n-2)}{m(n-2)^2(n-4)}$ for $n > 4$

- If $X \to F(m,n)$, then $\dfrac{1}{X} \to F(n,m)$.
- From the previous remark, we deduce that: $F_{1-\alpha}(m,n) = \dfrac{1}{F_\alpha(n,m)}$.

D/ The Exponential Family of Distributions (Class)

In this particular part of the course, we focus on **parametric probability distributions**, i.e., those that depend on a parameter $\theta \in \Theta \subset IR$ (or $\theta \in \Theta \subset IR^k$).
Let be the parametric family of distributions $\{f(x;\theta); \theta \in \Theta \subset IR\}$, where $f(x;\theta)$ represents the **probability density function** $f(x)$ that depends on the parameter θ for the continuous case, or the **probability mass function** $\mathrm{P}(x)$ that depends on the parameter θ for the discrete case.

<u>Definition 17</u>: A distribution $\{f(x;\theta); \theta \in \Theta \subset IR\}$ belongs to the **exponential family** if $f(x;\theta)$ it can be written in the following **canonical form**:

$$f(x;\theta) = a(\theta).b(x).\exp\{c(\theta).d(x)\}, \quad \forall x \in IR$$

If $\theta \in \Theta \subset IR^k$, the canonical form is written as:

$$f(x;\theta) = a(\theta).b(x).\exp\left\{\sum_{i=1}^{k} c_i(\theta).d_i(x)\right\}, \quad \forall x \in IR \ .$$

<u>Remark 23</u>: It is clear that if $X(\Omega)$ depends on the parameter θ, we cannot obtain the above canonical form.

<u>Remark 24</u>: In the case where θ has dimension k ($\theta \in \Theta \subset IR^k$), the canonical form given in the definition applies only to the **unknown parameters (of interest)**. For example, we will see that a given distribution $\mathrm{B}(n,\theta)$ belongs to the **exponential family** only if n is **known** and p is **unknown**. However, if n is **unknown**, then $\mathrm{B}(n,\theta)$ does **not** belong to the **exponential family**.

<u>Remark 25</u>: Parametric distributions that belong to the **exponential family** share many properties in **estimation theory** and **hypothesis testing theory**. Moreover, the functions $d_i(x)$ play an important role in **finding optimal estimators**.
[That's why it's so important to get to know the exponential family of distributions]

<u>Examples</u>:

- <u>The distribution</u> $\mathrm{B}(n,p)$ with n known:

  $$f(x;p) = C_n^x p^x (1-p)^{n-x}, \quad x = 0,1,\ldots,n$$
  $$= (1-p)^n.C_n^x \exp\left\{x.\log\left(\tfrac{p}{1-p}\right)\right\}$$

  It follows the canonical form given in **Definition 17**, with:

  $d(x) = x$, $c(p) = \log\left(\tfrac{p}{1-p}\right)$, $a(p) = (1-p)^n$, $b(x) = C_n^x$.

- The distribution $\mathrm{B}(p)$: $\theta = p \in]0,1[\subset IR$
  Since we know that $\mathrm{B}(p) = \mathrm{B}(1, p)$, it is sufficient to replace n with 1 in the previous example.
- The distribution $Poisson(\lambda)$: $\theta = \lambda \in IR_+^* \subset IR$
  $$f(x;\lambda) = e^{-\lambda} \tfrac{\lambda^x}{x!}, \quad x \in IN$$
  $$= e^{-\lambda}.\tfrac{1}{x!}.\exp\{x.\log(\lambda)\}$$
  It follows the canonical form given in **Definition 17**, with: $d(x) = x$, $c(\lambda) = \lambda$, $a(\lambda) = e^{-\lambda}$, $b(x) = \frac{1}{x!}$.
- The distribution $Exp(\lambda)$: $\theta = \lambda \in IR_+^* \subset IR$
  $$f(x;\lambda) = \lambda e^{-\lambda.x}.1_{IR^+}(x)$$
  It follows the canonical form given in **Definition 17**, with:
  $d(x) = x$, $c(\lambda) = \lambda$, $a(\lambda) = \lambda$, $b(x) = 1_{IR^+}(x)$
- The distribution $\Gamma(p,\lambda)$: $\theta = (p,\lambda) \in IR \times IR_+^* \subset IR^2$
  $$f(x;\lambda) = \tfrac{1}{\Gamma(p)}.\lambda^p.\exp\{-\lambda.x + (p-1)\log(x)\}.1_{IR^+}(x)$$
  It follows the canonical form given in **Definition 17**, with:
  $d_1(x) = x$, $d_2(x) = \log(x)$, $c_1(p,\lambda) = -\lambda$, $c_2(p,\lambda) = p-1$, $a(p,\lambda) = \frac{\lambda^p}{\Gamma(p)}$ et $b(x) = 1_{IR^+}(x)$.
  By setting $p = 1$, we retrieve the case of the distribution $Exp(\lambda)$ (since $Exp(\lambda) = \Gamma(1,\lambda)$).
- The distribution $N(\mu,\sigma)$: $\theta = (\mu,\sigma) \in IR \times IR_+^* \subset IR^2$
  $$f(x;\mu,\sigma) = \tfrac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\tfrac{1}{2\sigma^2}(x-\mu)^2\right\}, \quad x \in IR$$
  $$= \tfrac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\tfrac{\mu^2}{2\sigma^2}\right\}.\exp\left\{-\tfrac{1}{2\sigma^2}x^2 + \tfrac{\mu}{\sigma^2}x\right\}$$
  It follows the canonical form given in **Definition 17**, with:
  $d_1(x) = x^2$, $d_2(x) = x$, $c_1(\mu,\sigma) = -\frac{1}{2\sigma^2}$, $c_2(\mu,\sigma) = \frac{\mu}{\sigma^2}$, $a(\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{\mu^2}{2\sigma^2}\right\}$ et $b(x) = 1$.
- **Examples of Distributions that Do Not Belong to the Exponential Family,**
  Some distributions do **not** belong to the **exponential family**. It is easy to verify that:
  - The **Uniform Distribution** $U_{[0,\theta]}$ defined by the probability density function
    $f(x;\theta) = \frac{1}{\theta}.1_{[0,\theta]}(x)$
  - The **Cauchy Distribution** defined by the probability density function
    $f(x;\theta) = \frac{1}{\pi}.\frac{1}{1+(x-\theta)^2}.1_{IR}(x)$

  cannot be rewritten in the **canonical form** given in **Definition 17**, and therefore, they **do not** belong to the **exponential family**.